

Guillaume Jeanneret Sanmiguel

Curriculum Vitae

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About me

Summary Postdoctoral R&D Researcher in Computer Vision and Deep Learning at ISIR (Sorbonne Université) under Matthieu Cord, holding a PhD supervised by Frédéric Jurie and Loïc Simon, and a Master's research background with Pablo Arbeláez. Specialized in generative models (diffusion), explainable AI (XAI), and adversarial robustness, with a strong track record of designing SOTA algorithms published at top-tier venues (CVPR, ICCV, ECCV, WACV). Experienced in managing the end-to-end research lifecycle—from foundational math and PyTorch prototyping to large-scale model evaluation. Active reviewer for top AI conferences (CVPR, ICCV, TPAMI), trilingual in Spanish, English and French, and seeking an AI Research or Applied Scientist role to translate cutting-edge vision research into impactful technologies

Areas of Interest Deep Learning, Computer Vision, Explainable AI, Video Generation.

Nationalities Swiss and Colombian

Education

08/21–08/24 **PhD Computer Science**, *Unicaen, ENSICAEN, GREYC Laboratory*, Caen, France.

2018–2020 **M.Sc. in Biomedical Engineering**, *Universidad de los Andes, Bogotá*, Colombia.

2013–2018 **B.Sc. Biomedical Engineering**, *Universidad de los Andes, Bogotá*, Colombia.

2018 **Minor Mathematics and Computational Mathematics**, *Department of Mathematics*, Universidad de los Andes, Bogotá, Colombia.

2013 **Maturité Suisse Degree (Math and Physics emphasis)**, *Helvetia School*, Bogotá, Colombia.

2013 **High School Degree**, *Helvetia School*, Bogotá, Colombia.

Experience

11/24–11/26 Post Doctoral Researcher at ISIR Lab (Sorbonne Université), Paris, France

08/21–08/24 PhD in Computer Science (Université de Caen Normandie), Caen, France

01/18–07/21 Researcher at the Biomedical Computer Vision Group, Bogotá, Colombia.
01/19–12/20 Graduate Research Assistant (Universidad de los Andes, Bogotá, Colombia)
Summer 2017 MTBI – Research Experience for Undergraduates (Arizona State University, Arizona, USA)

Teaching Experience

09/25–01/26 Graduate Teaching Assistant: Pattern Recognition for Image Analysis and Interpretation (Sorbonne Université), Paris, France
10/23–12/23 Graduate Teaching Assistant: Tools for Computer Vision (ENSICAEN, Caen, France)
10/22–12/22 Graduate Teaching Assistant: Tools for Computer Vision (ENSICAEN, Caen, France)
07/18–12/18 Graduate Teaching Assistant: Image Processing and Analysis (Universidad de los Andes, Bogotá, Colombia)
01/18–06/18 Graduate Teaching Assistant: Scientific Programming (Universidad de los Andes, Bogotá, Colombia)
07/17–12/17 Undergraduate Teaching Assistant: Processing and Image Analysis (Universidad de los Andes, Bogotá, Colombia)
01/16–12/17 Undergraduate Teaching Assistant: Accompaniment Program (Universidad de los Andes, Bogotá, Colombia)
07/15–12/15 Undergraduate Teaching Assistant: Algorithm, and Object Oriented Programming (Universidad de los Andes, Bogotá, Colombia)

Published Papers

ECCVW 2026 Spatially-Grounded Text-to-Video Generation via Inference-Time Gradient-Free Optimization
ECCVW 2026 FrescoDiffusion: 4K Image-to-Video with Prior-Regularized Tiled Diffusion
CVPRW 2025 Disentangling Visual Transformers: Patch-level Interpretability for Image Classification
CVPRW 2025 OnlyFlow: Optical Flow based Motion Conditioning for Video Diffusion Models
WACV 2024 Text-to-Image Models for Counterfactual Explanations: a Black-Box Approach.
CVIU 2024 Diffusion Models for Counterfactual Explanations (Journal Version)
ICCVW 2023 BoDiffusion: Diffusing Sparse Observations for Full-Body Human Motion Synthesis.
CVPR 2023 Adversarial Counterfactual Visual Explanations.
ACCV 2022 Diffusion Models for Counterfactual Explanations.
Scientific Reports 2022 Smart Pooling: AI-powered COVID-19 testing.
ICCVW 2021 A Hierarchical Assessment of Adversarial Severity.
ICCVW 2021 Enhancing Adversarial Robustness via Test-time Transformation Ensembling.
CVIU 2021 MAIN: Multi Attention Instance Network.
ECCV 2020 Gabor Layers Enhance Network Robustness.

Research Projects and Preprints

- Ph.D Thesis Towards Explainable and Interpretable Deep Neural Networks
- M.Sc. Thesis MINT: Multi Instance Network, an Efficient Framework for Video Object Segmentation.
- MTBI Project The Dynamics of Math Anxiety as it is Transferred through Peer and Teacher Interactions.

Awards and Achievements

Outstanding Reviewer: CVPR25, ICCV25, CVPR26

Adversarial Robustness in the Real World ICCV 2021 Workshop - Best Paper Award was given to the paper titled *A Hierarchical Assessment of Adversarial Severity*

ACCV 2022 Oral Presentation: *Diffusion Models for Counterfactual Explanations*

Languages

- Spanish Native
- English Fluent
- French Fluent (Swiss *Maturité*)